

Signal Processing

Exercise 8 – z-transform – Solution

SoSe 2026

Question 1

The left hand side shows several pole-zero plots, corresponding amplitude responses $|H(e^{j\Omega})|$, which are not in the correct order, are shown on the right hand side (Fig. 1).

Assign the pole-zero plots to each corresponding magnitude response and specify the systems' filter characteristics.

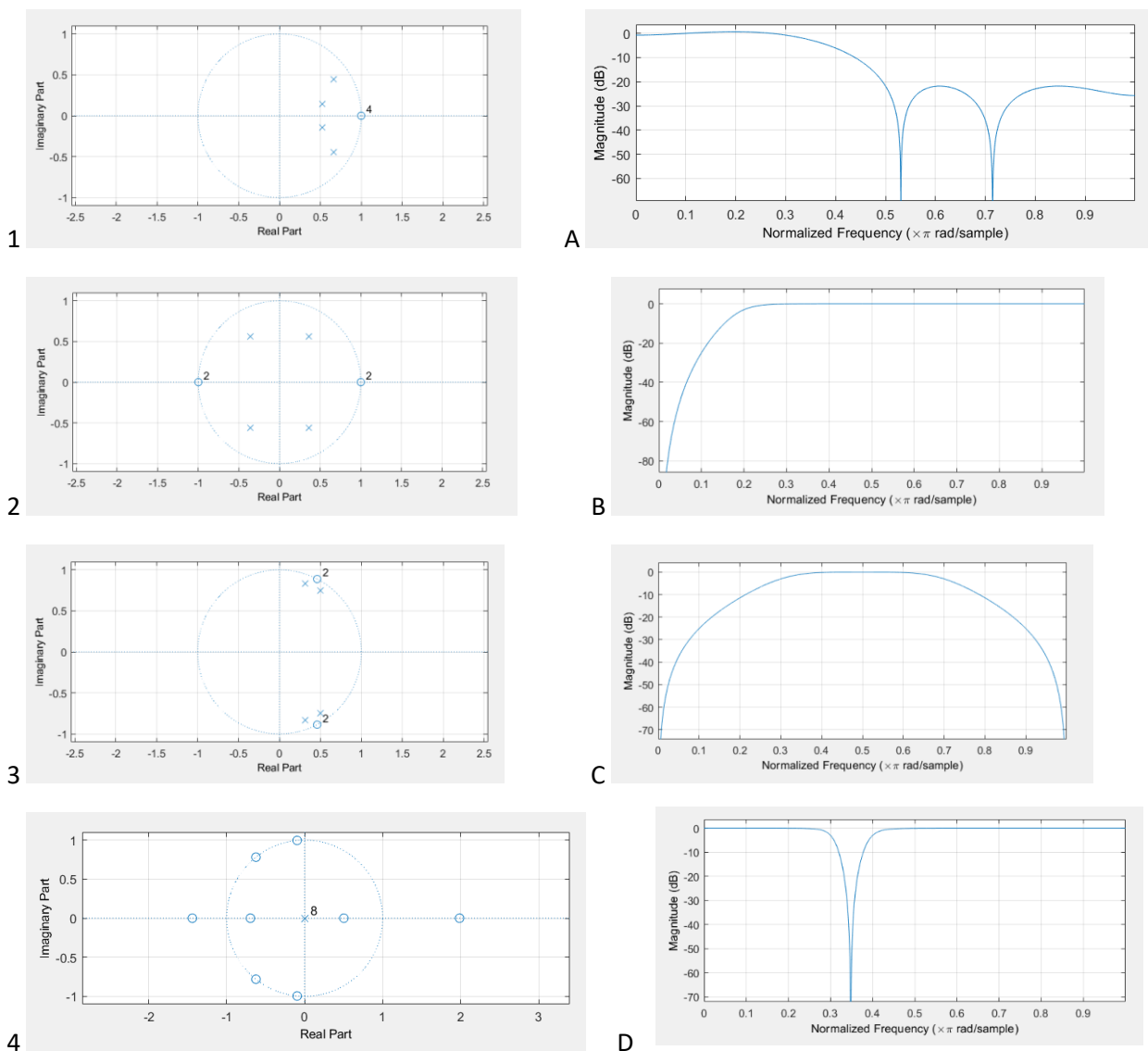


Figure 1: Pole-zero plot and corresponding, unsorted amplitude responses.

- | | | |
|-----|---|---|
| 1 → | B | Filter characteristic: Highpass |
| 2 → | C | Filter characteristic: Bandpass |
| 3 → | D | Filter characteristic: Bandstop / Notch |
| 4 → | A | Filter characteristic: Lowpass |

Question 2

A system is defined by the following difference equation:

$$y[n] = x[n] + x[n-1] - 0,1 \cdot y[n-1] + 0,3 \cdot y[n-2]$$

a) Determine the transfer function $H(z)$.

$$y[n] \text{ --- } Y(z)$$

$$Y(z) = X(z) \cdot z^0 + X(z) \cdot z^{-1} - 0,1 \cdot Y(z) \cdot z^{-1} + 0,3 \cdot Y(z) \cdot z^{-2}$$

$$Y(z) \cdot (1 + 0,1z^{-1} - 0,3z^{-2}) = X(z) \cdot (1 + z^{-1})$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{1 + z^{-1}}{1 + 0,1 \cdot z^{-1} - 0,3 \cdot z^{-2}} = \frac{z^2 + z}{z^2 + 0,1 \cdot z - 0,3}$$

b) Sketch the corresponding pole-zero plot and draw the region of convergence (ROC) (Fig. 2).

Zeros $z_{0,k}$: $z^2 + z = 0 \rightarrow z \cdot (z + 1) = 0$

$\rightarrow z_{0,1} = 0 ; z_{0,2} = -1$

Poles $z_{\infty,k}$: $z^2 + 0,1 \cdot z - 0,3$

$\rightarrow z_{\infty,1} = 0,5 ; z_{\infty,2} = -0,6$

ROC:

Causal system: ROC is an area outside a circle with radius $r = |z_{\infty,k}|_{\max}$

$\rightarrow r = |z_{\infty,2}| = 0,6$

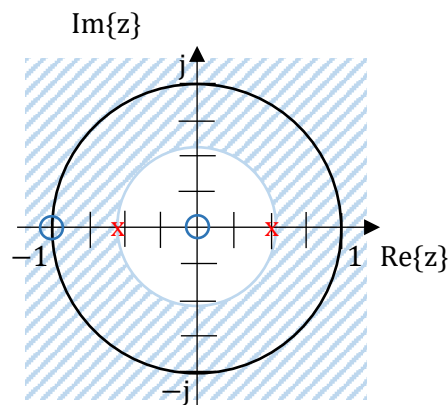


Figure 2: The system's pole-zero plot and its ROC.

Is the system stable, explain?

The system is stable, since the ROC includes the unit circle and all poles of the causal system are located inside the unit circle.

c) Is the system real / real valued?

Yes,

- all coefficients of the difference equation are real
- all the poles and zeros are real

HINT: Conjugate complex poles and zeros are also characteristic for real systems!

d) Determine the system's amplitude response $|H(e^{j\Omega})|$ and sketch it in the interval $\Omega = 0, \dots, 4\pi$ (Fig. 3). Define its filter characteristic.

$$H(z) = \frac{z \cdot (z + 1)}{(z - 0.5) \cdot (z + 0.6)}$$

Real system: Amplitude response is conjugate symmetric and it is 2π periodic

→ Evaluation for $\Omega = \{0, \frac{\pi}{2}, \pi\}$

Set $z = e^{j\Omega}$

$$H(e^{j\Omega}) = H(z)|_{z=e^{j\Omega}} = \frac{e^{j\Omega} \cdot (e^{j\Omega} + 1)}{(e^{j\Omega} - 0.5) \cdot (e^{j\Omega} + 0.6)}$$

I $\Omega = 0$

$$|H(e^{j0})| = \frac{e^{j0} \cdot (e^{j0} + 1)}{(e^{j0} - 0.5) \cdot (e^{j0} + 0.6)} = \frac{1 \cdot (1 + 1)}{(1 - 0.5) \cdot (1 + 0.6)} = 2.5$$

II $\Omega = \frac{\pi}{2}$

$$\begin{aligned} |H(e^{j\frac{\pi}{2}})| &= \frac{e^{j\frac{\pi}{2}} \cdot (e^{j\frac{\pi}{2}} + 1)}{(e^{j\frac{\pi}{2}} - 0.5) \cdot (e^{j\frac{\pi}{2}} + 0.6)} = \frac{|j \cdot (j + 1)|}{|(j - 0.5) \cdot (j + 0.6)|} = \frac{|-1 + 1j|}{|-1.3 + 0.1j|} \\ &= \frac{\sqrt{1^2 + 1^2}}{\sqrt{(-1.3)^2 + (0.1)^2}} \\ |H(e^{j\frac{\pi}{2}})| &\approx 1.1 \end{aligned}$$

Since the magnitude is calculated, we do not have to perform the complex conjugate extension! (Division of two complex numbers)

III $\Omega = \pi$

$$|H(e^{j\pi})| = \frac{-1 \cdot (-1 + 1)}{(-1 - 0.5) \cdot (-1 + 0.6)} = 0$$

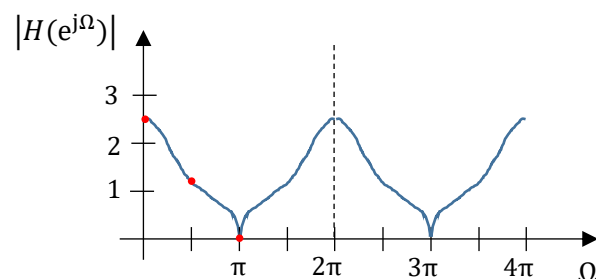


Figure 3: Amplitude response inside the interval $\Omega = 0, \dots, 4\pi$.

Question 3

A causal LTI system has a certain impulse response $h[n]$. Its transfer function $H(z)$ reads as

$$H(z) = \frac{1 + z^{-1}}{\left(1 - \frac{1}{2}z^{-1}\right) \cdot \left(1 + \frac{1}{4}z^{-1}\right)}$$

a) Please sketch the poles and zeros (Fig. 4).

$$H(z) = \frac{1 + z^{-1}}{\left(1 - \frac{1}{2}z^{-1}\right) \cdot \left(1 + \frac{1}{4}z^{-1}\right)} \cdot \frac{z^2}{z^2} = \frac{z(z + 1)}{\left(z - \frac{1}{2}\right) \cdot \left(z + \frac{1}{4}\right)}$$

Zeros: $z_{0,1} = 0$; $z_{0,2} = -1$

Poles: $z_{\infty,1} = 0.5$; $z_{\infty,2} = -0.25$

b) Draw the system's ROC (Fig. 4). Is the system stable, explain!

$$r = |z_{\infty,2}| = 0.5$$

The system is stable, since the ROC includes the unit circle and all poles of the causal system are located inside the unit circle.

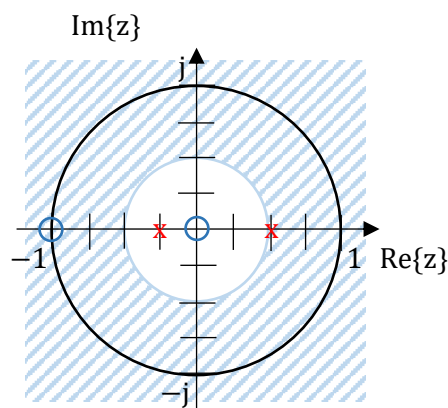


Figure 4: The system's pole-zero plot.

c) Determine the impulse response $h[n]$.

Partial fraction decomposition!

$$H(z) = \frac{1 + z^{-1}}{\left(1 - \frac{1}{2}z^{-1}\right) \cdot \left(1 + \frac{1}{4}z^{-1}\right)} = \frac{A}{\left(1 - \frac{1}{2}z^{-1}\right)} + \frac{B}{\left(1 + \frac{1}{4}z^{-1}\right)}$$

$$\frac{1 + z^{-1}}{\left(1 - \frac{1}{2}z^{-1}\right) \cdot \left(1 + \frac{1}{4}z^{-1}\right)} = \frac{A}{\left(1 - \frac{1}{2}z^{-1}\right)} \cdot \frac{\left(1 + \frac{1}{4}z^{-1}\right)}{\left(1 + \frac{1}{4}z^{-1}\right)} + \frac{B}{\left(1 + \frac{1}{4}z^{-1}\right)} \cdot \frac{\left(1 - \frac{1}{2}z^{-1}\right)}{\left(1 - \frac{1}{2}z^{-1}\right)}$$

$$1 + z^{-1} = A \cdot \left(1 + \frac{1}{4}z^{-1}\right) + B \cdot \left(1 - \frac{1}{2}z^{-1}\right)$$

Plug in poles: $z_{\infty,1} = 0.5 \rightarrow A = 2$; $z_{\infty,2} = -0.25 \rightarrow B = -1$

$$H(z) = \frac{2}{\left(1 - \frac{1}{2}z^{-1}\right)} + \frac{-1}{\left(1 + \frac{1}{4}z^{-1}\right)} = 2 \cdot \frac{z}{z - \frac{1}{2}} + (-1) \cdot \frac{z}{z + \frac{1}{4}}$$

$H(z) \bullet \text{---} \bigcirc h[n]$

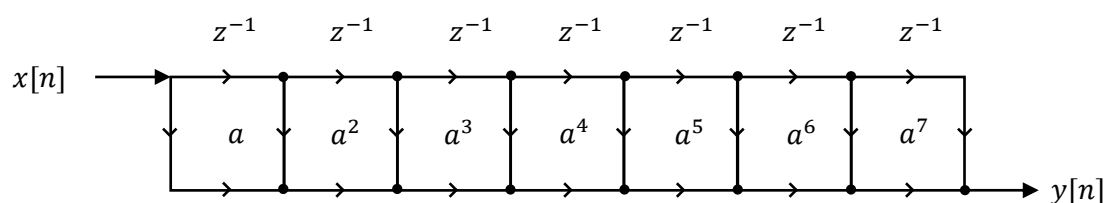
$$h[n] = 2 \cdot \frac{1^n}{2} \cdot \sigma[n] - \left(-\frac{1}{4}\right)^n \cdot \sigma[n]$$

Question 4

The impulse response $h[n]$ of a causal LTI system is defined

$$h[n] = \begin{cases} a^n, & \text{for } 0 \leq n \leq 7 \\ 0, & \text{otherwise} \end{cases}$$

a) Draw the signal flow graph of the system in direct form (non-recursive implementation).



b) Show that the corresponding transfer function can be expressed as

$$H(z) = \frac{1 - a^8 z^{-8}}{1 - a z^{-1}}, \quad |z| > |a|$$

Explanation:

Sum of a finite geometric series of q^n is

$$\sum_{n=0}^N q^n = \frac{1 - q^{N+1}}{1 - q}$$

$$H(z) = \sum_{n=0}^7 a^n z^{-n} = \sum_{n=0}^7 (a z^{-1})^n = \frac{1 - a^8 z^{-8}}{1 - a z^{-1}}$$

Question 5

Consider a time-discrete LTI system that is described by the following difference equation:

$$y[n] = x[n] - 2x[n-1] + \frac{5}{2}y[n-1] - y[n-2]$$

a) Determine the transfer function $H(z)$ in the frequency domain.

$$Y(z) = X(z) - 2X(z) \cdot z^{-1} + \frac{5}{2}Y(z) \cdot z^{-1} - Y(z) \cdot z^{-2}$$

$$Y(z) \cdot \left(1 - \frac{5}{2}z^{-1} + z^{-2}\right) = X(z) \cdot (1 - 2z^{-1})$$

$$H(z) = \frac{z^2 - 2z}{z^2 - \frac{5}{2}z + 1}$$

b) Determine the impulse response $h[n]$.

Calculate poles and apply partial fraction decomposition

Poles: $z_{\infty,1} = 2$; $z_{\infty,2} = 0.5$

Rewrite:

$$H(z) = \frac{z(z-2)}{(z-2) \cdot \left(z - \frac{1}{2}\right)} = z \cdot \left(\frac{A}{(z-2)} + \frac{B}{\left(z - \frac{1}{2}\right)} \right)$$

$$\frac{z(z-2)}{(z-2) \cdot \left(z - \frac{1}{2}\right)} = z \cdot \left(\frac{A \cdot \left(z - \frac{1}{2}\right)}{\left(z - \frac{1}{2}\right) \cdot \left(z - \frac{1}{2}\right)} + \frac{B \cdot (z-2)}{\left(z - \frac{1}{2}\right) \cdot (z-2)} \right)$$

$$z - 2 = A \cdot z - \frac{1}{2} \cdot A + b \cdot z - 2 \cdot b$$

Plug in poles: $z_{\infty,1} = 2 \rightarrow A = 0$; $z_{\infty,2} = 0.5 \rightarrow B = 1$

$$H(z) = z \cdot \left(\frac{0}{(z-2)} + \frac{1}{\left(z - \frac{1}{2}\right)} \right) = \frac{z}{z - \frac{1}{2}} \quad \bullet \text{---} \circ \quad h[n] = \left(\frac{1}{2}\right)^n \sigma[n]$$

c) Is the system stable? Please give a short explanation.

Yes, the one remaining pole of the causal system is located within the unit circle.

Question 6

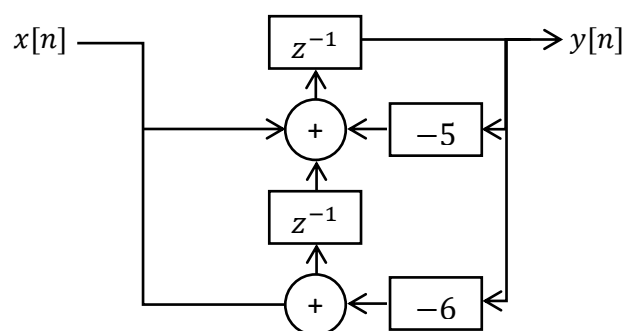
A time-discrete, causal system is defined by the following transfer function $H(z)$

$$H(z) = \frac{z+1}{z^2+5z+6}$$

a) Determine the difference equation $y[n]$ of the system.

$$y[n] = x[n-1] + x[n-2] - 6y[n-2] - 5y[n-1]$$

b) Draw the system's block diagram with respect to a minimum order of delay elements.



c) Please sketch the poles and zeros (Fig. 5). Is the system stable, explain!

Zeros: $z_{0,1} = -1$

Poles: $z_{\infty,1} = -2$; $z_{\infty,2} = -3$

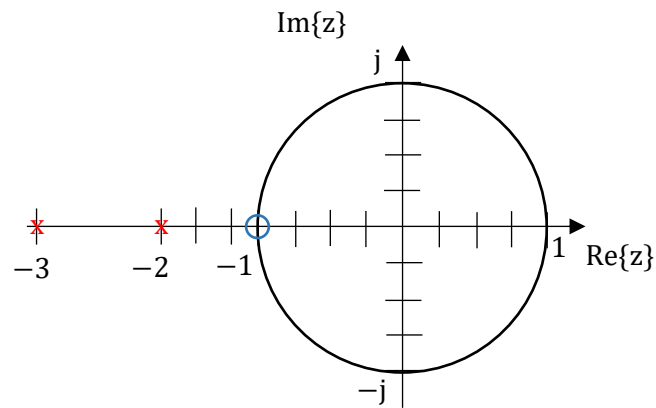


Figure 5: The system's pole-zero plot.

Causal system and poles are lying outside the unit circle → Not stable

d) Calculate the first five values of the system response $y[n]$ for $x[n] = \delta[n]$.

$$y[n] = \delta[n-1] + \delta[n-2] - 5y[n-1] - 6y[n-2]$$

$$n = 0$$

$$y[0] = \delta[-1] + \delta[-2] - 5y[-1] - 6y[-2]$$

$$y[0] = 0$$

$$n = 1$$

$$y[1] = \delta[0] + \delta[-1] - 5y[0] - 6y[-1]$$

$$y[1] = 1$$

$$n = 2$$

$$y[2] = \delta[1] + \delta[0] - 5y[1] - 6y[0]$$

$$y[2] = -4$$

$$n = 3$$

$$y[3] = 14$$

$$n = 4$$

$$y[4] = 4$$

e) Calculate the first five values of the system response $y[n]$ for $x[n] = 2\sigma[n]$.

$$n = 0$$

$$y[0] = 2\sigma[-1] + 2\sigma[-2] - 5y[-1] - 6y[-2]$$

$$y[0] = 0$$

$$n = 1$$

$$y[1] = 2$$

$$n = 2$$

$$y[2] = -6$$

$$n = 3$$

$$y[3] = 22$$

$$n = 4$$

$$y[4] = -70$$

Question 7

Consider the following block diagrams (Fig. 6).

Show that both diagrams represent the same discrete system.

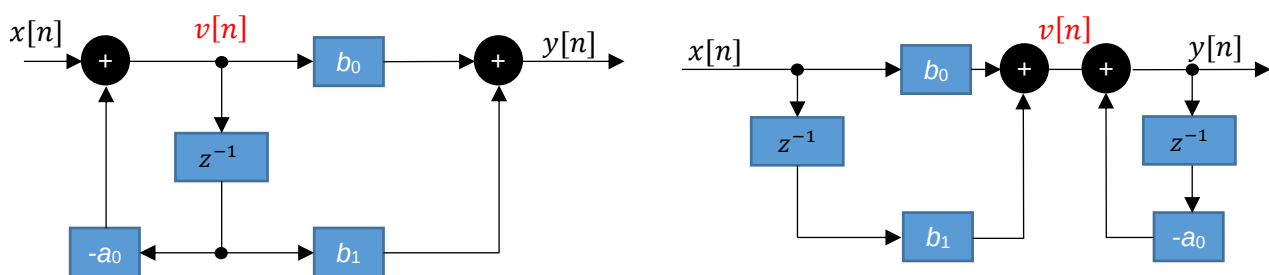


Figure 6: Block diagrams of the same discrete system.

System 1 (left)

Introduce intermediate step signal $v[n]$.

$$\text{I } v[n] = x[n] - a_0 \cdot v[n-1]$$

$$\text{II } y[n] = b_0 \cdot v[n] + b_1 \cdot v[n-1]$$

Apply z-transform for Eq. I and Eq. II

$$\text{I}' \quad V(z) = \frac{X(z)}{(1+a_0z^{-1})}$$

$$\text{II}' \quad Y(z) = b_0 \cdot V(z) + b_1z^{-1}V(z)$$

Plug I' in II' and calculate $H(z)$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{b_0z + b_1}{z + a_0}$$

System 2 (right)

Introduce intermediate step signal $v[n]$.

$$\text{I} \quad v[n] = x[n] \cdot b_0 + x[n-1] \cdot b_1$$

$$\text{II} \quad y[n] = v[n] - a_0 \cdot y[n-1]$$

Plug I in II and perform z-transform, then calculate $H(z)$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{b_0z + b_1}{z + a_0}$$